Electrical power and energy

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Learning target: Calculate electrical transfer rate and energy use.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 7, Day 8; 3 course days later.*
0.8 A, by charge conservation.

02. Explain a worked example (worked)
6 V, 2 A, 12 W and 5 s lead to 60 J; quantities are related but not interchangeable.

03. Look and explain (model)
The faster transfer has twice the average power.

04. Build the solution (practice)
12 W; 60 J.

05. Use the idea (reason)
10 W for 1 s=10 J; 5 W for 10 s=50 J. Duration matters.

06. Choose a method (mixed) *Review: Unit 6, Day 17; 12 course days later.*
The farther target takes 4 times as long because t=d/v.

07. Exit check (exit)
30 J.

08. STRETCH: Apply it another way (stretch)
6 and 3 s.

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01. Bring back an earlier idea (retrieve) *Review: Unit 7, Day 8; 3 course days later.*
Parallel 5 Ω; series 20 Ω.

02. Explain a worked example (worked)
6 V, 2 A, 12 W and 5 s lead to 60 J; quantities are related but not interchangeable.

03. Represent and interpret (model)
The faster transfer has twice the average power.

04. Solve and justify (practice)
9 W; 2160 J.

05. Analyze the reasoning (reason)
10 W for 1 s=10 J; 5 W for 10 s=50 J. Duration matters.

06. Choose a method (mixed) *Review: Unit 6, Day 17; 12 course days later.*
v≈20 m/s; t≈300 s. Requires the long-wave shallow-water regime and simplified depth; not a damage forecast.

07. Exit check (exit)
V=12 V; IV=24 W and $I^2 R$=24 W.

08. STRETCH: Apply it another way (stretch)
Input 720 J; useful power 10.8 W; efficiency 60%.